🌐 Module 3: Network Protocols & Communication (10.27.2026)

📜 The Rules of Communication

All communication has **3 elements**:

- 📤 Message source (sender)
- 📥 Message destination (receiver)
- 🔗 Channel

Protocols Must Include:

- ✅ Identified sender & receiver
- 🗣️ Common language & grammar
- ⏱️ Speed & timing of delivery
- 📬 Confirmation/acknowledgment requirements

Key Concepts:

- **Encoding** 🔄 → Converting info into transmittable form
- **Decoding** 🔄 → Reversing encoding to interpret info
- **Message Timing** ⏰ → Flow control, response timeout, access method
- **Delivery Options** 📮 → Unicast (1:1), Multicast (1:many), Broadcast (1:all)

🔧 Protocols

Implemented by **end devices** & **intermediary devices** (software/hardware/both)

Protocol Categories:

Function	Examples
🔐 Security	SSH, SSL, TLS
🛣️ Routing	OSPF, BGP
🔍 Auto-detection	DHCP, DNS

Protocol Functions:

- 📍 Addressing
- ✔️ Reliability
- 🚦 Flow control
- 🔢 Sequencing
- ⚠️ Error-detection
- 🔌 Application interface

📚 Protocol Suites

Definition: Group of inter-related protocols for communication functions

Protocol Stack 📊 → Shows how protocols within a suite are implemented

TCP/IP Suite:

- 🏆 Used by today's networks & internet
- 📖 Open standard
- 🏛️ Standards-based

Layers covered: Application, Transport, Internet

🏛️ Standards Organizations

Organization	Role
🌍 ISOC, IAB, IETF, IRTF	Internet standards
📋 ICANN, IANA	TCP/IP support
⚡ IEEE, EIA, TIA, ITU-T	Electronic/communications

> 💡 **Open standards** encourage interoperability, competition & innovation

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📐 Reference Models

OSI Model (7 Layers)

```
7️⃣ Application
6️⃣ Presentation
5️⃣ Session
4️⃣ Transport
3️⃣ Network
2️⃣ Data Link
1️⃣ Physical
```

TCP/IP Model (4 Layers)

```
4️⃣ Application
3️⃣ Transport
2️⃣ Internet
1️⃣ Network Access
```

📦 Data Encapsulation

Benefits of Segmentation:

1. 🔀 **Multiplexing** → Multiple conversations interleaved
2. ⚡ **Efficiency** → Only missing parts retransmitted

Key Terms:

- **PDU** (Protocol Data Unit) → Data form at each layer
- **Encapsulation** ⬇️ → Top-to-bottom process (sending)
- **De-encapsulation** ⬆️ → Bottom-to-top process (receiving)

> 🔑 TCP handles sequencing of segments

🎯 Data Access

Layer Responsibilities:

Layer	Address Purpose
🌐 Network Layer	Source → Final destination (IP)
🔗 Data Link Layer	NIC → NIC (same network)

IP Address Components:

- **Network portion** (IPv4) / **Prefix** (IPv6)
- **Host portion** (IPv4) / **Interface ID** (IPv6)

Delivery Scenarios:

- 🏠 **Same network** → Frame sent directly to receiver
- 🌍 **Different network** → Frame sent to **router/default gateway**

Ethernet Addresses:

- Known as **MAC addresses** 🖨️
- Used for NIC-to-NIC delivery on same network